



SurakshitPath (सुरक्षितपथ)

VIDEO DEMO SCRIPT & FEATURE GUIDE

AI-Powered Night-Safety Navigation & Municipal Dark-Spot Audit Platform

TOTAL DURATION
2 Minutes 45 Seconds

TESTBED REGION
Pune (Tathawade ⇄ Kothrud)

SDG ALIGNMENT
UN SDG 5 & SDG 11

CARTOGRAPHY
Google Maps Satellite Hybrid

Scene 1: The Nocturnal Problem Hook

0:00 - 0:25 (25s)

ON-SCREEN ACTION

- Start on **http://localhost:5173/** in full screen (F11).
- Display the high-resolution Google Maps Satellite view over Pune.
- Pan across the Tathawade-Kothrud urban corridor.

SPOKEN VOICEOVER

"Every night, millions of women, students, and workers in our cities face an invisible dilemma: standard navigation apps blindly recommend the shortest route—even if it leads through pitch-black alleys, unlit underpasses, and isolated service lanes just to save two minutes. Welcome to SurakshitPath—an AI-powered safe-routing and civic audit platform built to make night mobility predictable, transparent, and safe."

Core Feature Highlighted: Problem Statement, Nocturnal Mobility Gap, and Live Satellite Hybrid Cartography.

Scene 2: Google Maps Style Search & Dual-Route Model

0:25 - 0:55 (30s)

ON-SCREEN ACTION

- Show the 3-column Directions Box with travel mode tabs (Safe, Walk, Drive, Transit).
- Point out Origin (JSPM Tathawade) & Destination (Kothrud Stand).
- Click between **Route 2 (Recommended Safe Corridor)** and **Route 1 (Direct Cut - Fastest)**.
- Highlight the circular **Safety Score Gauge (72 vs 45)** and green corridor line on the map.

SPOKEN VOICEOVER

"SurakshitPath features an intuitive Google Maps-style directions interface. When planning a trip from Tathawade to Kothrud, our graph routing engine doesn't just calculate distance—it evaluates street illumination, 24/7 commercial frontage, CCTV presence, and emergency shelter proximity. Look at the comparison: the direct route is 3 minutes faster, but exposes commuters to four unlit high-risk cuts. SurakshitPath highlights the Recommended Safe Corridor with 98% continuous lighting and 4 verified safe shelters."

Core Feature Highlighted: Multi-Attribute Dijkstra Dual-Route Model, Circular Radial Score Gauge, and Safe Haven Counts.

Scene 3: Commuter-Friendly Safety Preference & Emergency Shelter Lock

0:55 - 1:25
(30s)

 ON-SCREEN ACTION

- Expand the **Route Safety Priority** accordion.
- Slide the slider across: *Speed* → *Balanced* → *Max Safe*.
- Show how the live status badge changes to '  Maximum Protection'.
- Click the **1-Click Emergency Shelter Lock** card to instantly route to nearest 24/7 hospital/pharmacy.

 SPOKEN VOICEOVER

"Commuters can customize their safety preference with our user-friendly priority slider—from 'Speed First' for daylight shortcuts to 'Maximum Safe Corridor' which strictly stays on illuminated arterial highways. In any urgent situation, the 1-Click Emergency Shelter Lock instantly pinpoints and reroutes to the nearest verified 24/7 hospital, pharmacy, or police chowki with live distance metrics."

Core Feature Highlighted: User-Understandable Safety Slider, Beta Impedance Weighting, and Instant Emergency Safe Haven Lock.

Scene 4: Explainable AI (XAI) & TreeSHAP Local Attributions

1:25 - 1:55 (30s)

ON-SCREEN ACTION

- Expand **AI Explainability & SHAP Risk**.
- Show the animated **SVG 4-Factor Dimension Radar** (Illumination, Public Presence, Transit, Emergency Reach).
- Toggle to **TreeSHAP (ML)** to display game-theoretic positive and negative attribution bars.

SPOKEN VOICEOVER

"Unlike black-box navigation algorithms, SurakshitPath provides full explainability. Our 4-Factor Dimension Radar charts the exact safety attributes of your corridor, while our integrated TreeSHAP machine learning model provides game-theoretic feature attributions—explaining exactly why an edge was penalized or promoted relative to the base safety expectation."

Core Feature Highlighted: Explainable AI (XAI), SVG Polygon Radar Visualization, and TreeSHAP Decision Boundary Analysis.

Scene 5: Live Telemetry Watchdog & Anomaly Detection (Climax)

1:55 - 2:25 (30s)

ON-SCREEN ACTION

- Click ⚡ **Simulation & Watchdog** → Click **Start Simulated Trip**.
- Watch the pulsing commuter marker move along the safe path.
- Click **'Stray Off-Route (>50m)'**.
- Show the unskippable **'Are You Safe?'** modal with countdown timer and audio beep alert.
- Click **'Dispatch SOS Alert'**.

SPOKEN VOICEOVER

"During transit, our real-time Telemetry Watchdog with Kalman Filter smoothing tracks the commuter along the safe corridor envelope. If a commuter strays more than 50 meters off-route into an unmonitored lane or experiences a prolonged stall in an isolated stretch, the system triggers an unskippable safety check. If unacknowledged within 45 seconds, an automated emergency SOS is dispatched with live GPS coordinates to guardians and police."

Core Feature Highlighted: Telemetry Watchdog, Kalman Filter Geofencing, Cross-Track Distance ($d_{\perp}$), and Auto SOS Dispatch.

Scene 6: Multi-Role Civic Ecosystem & Municipal Audit

2:25 - 2:45 (20s)

ON-SCREEN ACTION

- Switch Role dropdown to **Guardian** (companion live radar).
- Switch to **Volunteer** (proximity SOS responder cards).
- Switch to **Civic Admin** (PMC/PCMC dark-spot SCADA table & Export JSON button).

SPOKEN VOICEOVER

"SurakshitPath bridges citizens with civic authorities. Guardians monitor trip progress and phone battery; volunteers receive proximity alerts to assist commuters; and municipal corporations (PMC & PCMC) receive prioritized SCADA dark-spot audit reports to dispatch streetlight repair work orders directly where safety risk is highest."

Core Feature Highlighted: Guardian Live Companion, Volunteer Responder Network, and Municipal Dark-Spot Audit Export.

 Comprehensive Feature Breakdown Matrix

Module / Feature	Everyday Commuter Benefit	Technical / Algorithmic Engine
Dual-Route Comparison	Compares fastest shortcut vs. well-lit safe corridor with clear duration and safety scores.	Multi-Attribute Graph Routing with edge impedance penalties $C(e) = \text{Length} \times (1 + \beta \cdot (1 - S_e))$.
Google Maps Satellite Hybrid	High-resolution satellite terrain with overlaid road names and landmark labels.	Leaflet Satellite Hybrid tile mapping with zero watermarking and custom route glow polyline styling.
Route Safety Priority	User-friendly slider (Speed, Balanced, Safe, Max Safe) to control detour tolerance.	Dynamic β sensitivity multiplier (0.0 to 1.5) regulating virtual length cost scaling.
1-Click Emergency Shelter Lock	Instantly calculates route to closest 24/7 pharmacy, hospital, or police post.	Euclidean nearest-neighbor search over curated Pune safe haven GeoJSON database.
Explainable AI & TreeSHAP	Explains exactly why a road is safe (lighting, commercial frontage, transit proximity).	Local SHAP waterfall additive attributions + 4-factor concentric polygon radar chart.
Telemetry Watchdog & Anomaly SOS	Monitors for route deviation (>50m) and stalls, triggering unskippable SOS countdown.	Kalman Filter state estimation, cross-track distance watchdog ($d_{\perp}$), and Web Audio API alerts.
Multi-Role Civic Ecosystem	Dedicated cockpits for Commuters, Guardians, Volunteer Responders, and Civic Admins.	Role-based UI state architecture with URL query parameter synchronization and JSON audit report export.

 Pro-Tips for Demo Recording:

- **Resolution & Zoom:** Record at 1920×1080 (1080p) or 4K. Use Chrome in full screen (F11) with browser zoom at 100%.
- **Pacing:** Pause for 1–2 seconds after each action (e.g. after clicking 'Stray Off-Route' or 'Reroute') so judges can absorb the visual state.
- **Live Audio:** Ensure your system audio is captured to hear the gentle anomaly beep alert when the Safety Check modal appears.